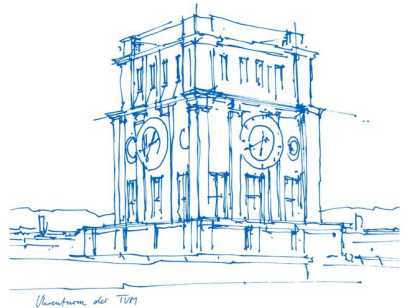


FineFlow–Espresso: A Calibratable 1D Model of Fines Migration

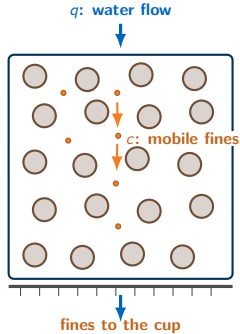
Connecting CT structure, pore-network
physics and hydraulic measurements

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Key variables describe flow, fines and pore structure



Flow and mobile fines

- Q : flow rate; A : puck area; $q = Q/A$: superficial velocity;
- c : mobile-fines concentration; $q c$: advective fines flux.

Pore structure and retained fines

- ε : porosity; K : permeability;
- m_a : available attached fines; m_d : deposited fines;
- d_f : fine diameter; d_h : local hydraulic diameter.

The goal is a model that measurements can actually identify

How can we predict flow, blockage and fines in the cup without hiding the physics inside fitted parameters?

CT images

Porosity, hydraulic diameter,
deposit location and axial
heterogeneity

Pore networks

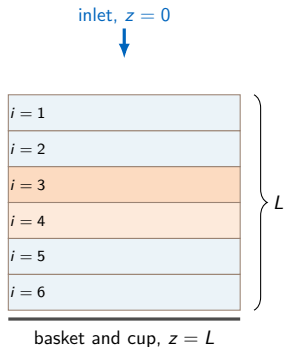
Permeability, dispersion,
particle capture and escape
closures

Experiments

Pressure–flow curves, pump
dynamics and time-resolved
cup fines

The equation structure is fixed first; each parameter is assigned to a specific measurement or pore-scale calculation.

One-dimensional means “layers in series”



Resolved along the puck height

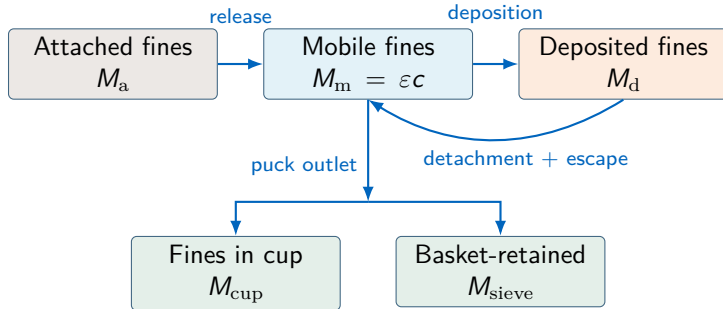
- pressure gradient;
- porosity and permeability;
- available, mobile and deposited fines.

Not resolved

- radial channels;
- individual pores or particles;
- extraction chemistry.

The same flow crosses every layer; a blocked layer carries a larger pressure gradient.

The model follows fines through five inventories



$$M_0 = M_a + M_m + M_d + M_{\text{sieve}} + M_{\text{cup}}.$$

Every fine is accounted for; internal exchange terms cancel in the global mass balance.

Three balance equations describe the evolving fines

$$\frac{\partial(\varepsilon c)}{\partial t} + \frac{\partial(q c)}{\partial z} = \frac{\partial}{\partial z} \left(\varepsilon D \frac{\partial c}{\partial z} \right) + R_{\text{rel}} - R_{\text{dep}} + R_{\text{wash}},$$

$$\frac{\partial m_d}{\partial t} = R_{\text{dep}} - R_{\text{wash}},$$

$$\frac{\partial m_a}{\partial t} = -R_{\text{rel}}.$$

$q c$

advection carries fines with
the water

$\varepsilon D \partial c / \partial z$

dispersion spreads the
concentration

R terms

local transfer between
inventories

Only the mobile fines travel between cells; release, deposition and washout are local source terms.

Release and deposition describe two opposite transfers

Release: fines enter the water

$$R_{\text{rel}} = k_{\text{rel}} m_{\text{a}} \left(\frac{|q|}{q_{\text{ref}}} \right)^{a_{\text{r}}}.$$

Available fines m_{a} are released into the flowing water. The release becomes faster with increasing flow; k_{rel} sets the timescale and a_{r} the flow sensitivity.

Calibrate using fines breakthrough at different flow rates.

Deposition: mobile fines are captured

$$R_{\text{dep}} = k_{\text{dep}} \chi(z) |q| c \left[1 - \frac{m_{\text{d}}}{m_{\text{cap}}} \right]_{+}.$$

The transported fines $|q|c$ are captured by the pore structure. Capture decreases as m_{d} approaches the local capacity m_{cap} . The optional factor $\chi(z)$ represents enhanced deposition near the basket.

Calibrate using PNM breakthrough and CT deposition profiles.

Release adds fines to the water; deposition stores them in the puck.

Washout requires detachment and geometric escape

Step 1: hydraulic detachment

$$R_{\text{det}}^* = k_{\text{det}} m_d \left[\frac{G}{G_{\text{crit}}} - 1 \right]_+^b,$$

$$G = \left| \frac{\partial p}{\partial z} \right|.$$

Deposits detach only when the pressure gradient exceeds G_{crit} . Above this threshold, stronger loading produces faster detachment.

Calibrate using pressure-pulse experiments.

Step 2: geometric escape

$$\lambda = \frac{d_f}{d_h}, \quad R_{\text{wash}} = P_{\text{esc}}(\lambda) R_{\text{det}}^*,$$

$$P_{\text{esc}} = \frac{1}{1 + \exp[s(\lambda - \lambda_c)]}.$$

A detached fine escapes only if it is small enough relative to the available pore size: small λ means easier escape.

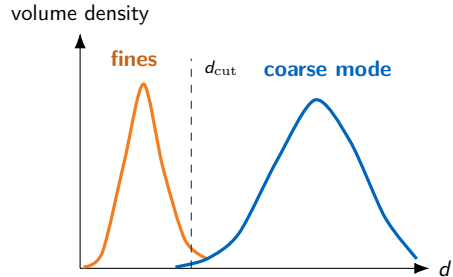
CT provides d_h , the PSD provides d_f , and PNM calibrates P_{esc} .

Detachment breaks adhesion; geometric escape determines actual washout.

The PSD is reduced to quantities used by the 1D closures

Measured bimodal PSD

- Fit a five-parameter mixture model.
- Choose a mobility cutoff d_{cut} .
- Integrate the fine fraction below the cutoff.
- Calculate representative fine and coarse diameters.



$$f_{\text{fine}} \rightarrow m_{a,0}, \quad d_f \rightarrow d_f/d_h, \quad d_{\text{eff}} \rightarrow d_{h,0} \text{ and Ergun baseline.}$$

Source: Bora and Briesen, Journal of Food Process Engineering 49 (2026), e70645.

Deposits change porosity and permeability

Porosity decreases

$$\varepsilon = \max\left(\varepsilon_0 - \frac{m_d}{\rho_d}, \varepsilon_{\min}\right).$$

Permeability controls resistance

$$\frac{K}{K_0} = \exp\left(-\beta \frac{m_d}{m_{\text{cap}}}\right)$$

with a configured lower bound.

Three selectable closures

- exponential deposit law;
- Kozeny–Carman-style porosity law;
- tabulated $m_d \mapsto K/K_0$ from PNM.

The PNM table is preferred when reconstructed pore networks are available: it preserves the simulated relation rather than forcing it into one empirical equation.

CT and PNM determine the initial structure; repeated PNM calculations determine how blockage changes conductance.

Darcy–Forchheimer converts structure into pressure loss

What pressure gradient is needed to drive the common superficial velocity $q = Q/A$?

$$G(z, t) = \underbrace{\frac{\mu q}{K(z, t)}}_{\text{viscous Darcy term}} + \underbrace{\rho \beta_F(z, t) |q| q}_{\text{nonlinear inertial term}}.$$

$$\Delta p_{\text{puck}} = q \int_0^L \frac{\mu}{K} dz + q^2 \int_0^L \rho \beta_F dz \quad (q \geq 0).$$

K : permeability from CT–PNM or low-flow measurements; β_F : Forchheimer coefficient from PNM or curvature in $\Delta p(Q)$.

The model does not solve a momentum PDE; it evaluates and integrates this local resistance law.

Why not use the Ergun equation directly?

Darcy only

Uses structural permeability K , but assumes pressure loss is linear in flow.

Best for sufficiently slow flow.

Direct Ergun

Predicts both terms from porosity, sphericity and one effective diameter.

Useful baseline, but compresses connectivity and bottlenecks into bulk descriptors.

Darcy–Forchheimer

Keeps CT–PNM K and adds a separately calibratable inertial coefficient β_F .

Selected formulation.

$$\beta_{F,\text{Ergun}} = \frac{1.75(1 - \varepsilon)}{\varepsilon^3 d_{\text{eff}}} \quad \text{provides an initial comparison, not a mandatory closure.}$$

We retain Ergun's nonlinear physics without discarding the permeability resolved from the real pore structure.

The basket and machine are separate from the coffee puck

Simple basket model

$$\Delta p_{\text{total}} = \Delta p_{\text{puck}} + QR_s,$$

$$\dot{M}_{\text{cup}} = (1 - \eta_s) \dot{M}_{\text{out,puck}},$$

$$\dot{M}_{\text{sieve}} = \eta_s \dot{M}_{\text{out,puck}}.$$

R_s and η_s are constant. Basket deposits do not yet alter resistance.

PI-controlled pump

$$e = p_{\text{set}} - p,$$

$$Q_{\text{raw}} = Q_{\text{ff}} + K_P e + K_I \int e dt,$$

$$Q_{\text{cmd}} = \text{clip}(Q_{\text{raw}}, 0, Q_{\text{max}}).$$

Pump lag and anti-windup are included. A coarse permeable puck can remain below 9 bar when Q_{max} is reached.

Empty-basket tests characterize R_s and the machine before any puck kinetics are fitted.

Initial and boundary conditions complete the mathematical problem



At $t = 0$

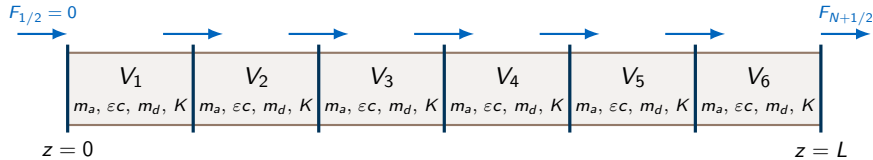
- $m_a = m_{a,0}$;
- $m_d = m_{d,0}$;
- $c = c_0$;
- $\varepsilon_0(z)$ and $K_0(z)$ are uniform or supplied as CT-PNM profiles.

At the boundaries

- clean water enters: $(q c)_{z=0} = 0$;
- zero dispersive flux at $z = 0, L$;
- mobile fines leave convectively:
 $\dot{M}_{\text{out}} = A q c(L, t)$;
- cup pressure is the zero-gauge reference.

Reverse flow and radial bypass channels are outside the present 1D model.

The finite-volume method enforces flux balance cell by cell



$$\frac{(\varepsilon c)_i^{n+1} - (\varepsilon c)_i^n}{\Delta t} = -\frac{F_{i+1/2} - F_{i-1/2}}{\Delta z} + R_{\text{rel},i} - R_{\text{dep},i} + R_{\text{wash},i}.$$

$$F_{i+1/2} = \underbrace{q c_i}_{\text{advection}} - \underbrace{\varepsilon_{i+1/2} D \frac{c_{i+1} - c_i}{\Delta z}}_{\text{dispersion}}, \quad q \geq 0.$$

$F_{i+1/2}$ is the total fines-mass flux through the face between cells i and $i + 1$. Upwind advection uses the concentration in the upstream cell, while centred dispersion uses the concentration difference between both cells.

Velocity and pressure are calculated algebraically

One common superficial velocity

$$\frac{\partial q}{\partial z} = 0 \quad \implies \quad q(t) = \frac{Q(t)}{A}.$$

The flow rate is determined by the selected operating mode:

- prescribed directly in flow mode;
- obtained from the pressure condition in pressure mode;
- determined by the pump and PI controller in PI mode.

The local pore velocity is then

$$u_{\text{pore},i} = \frac{q}{\varepsilon_i}.$$

Pressure is reconstructed cell by cell

$$G_i = \frac{\mu q}{K_i} + \rho \beta_{F,i} |q| q,$$

$$\Delta p_i = G_i \Delta z.$$

Starting from the pressure above the basket,

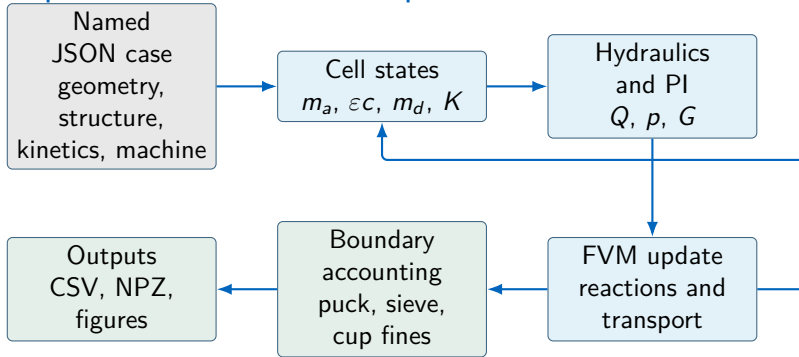
$$p_{N+1/2} = QR_s,$$

the pressure is accumulated towards the inlet:

$$p_{i-1/2} = p_{i+1/2} + G_i \Delta z.$$

A cell with lower permeability therefore produces a larger local pressure drop.

Each time step closes a feedback loop



Deposit changes ε , K and d_h ; these change pressure and flow; the new hydraulics then alter release, deposition and washout.

The coupling is dynamic even though the hydraulic relation itself is algebraic.

Calibration starts with structure and hydraulics

Category	Model inputs	How they are obtained
Measured and fixed	L, A, μ, ρ , dose, five PSD parameters	Geometry and temperature measurements; laser-diffraction PSD fit; calculate fines fraction and representative diameters.
CT-derived structure	$\varepsilon_0(z), d_{h,0}(z)$, deposit location	Segment axial slabs; calculate connected porosity and $d_h = 4V_{\text{void}}/A_{\text{wetted}}$; quantify segmentation uncertainty.
PNM-derived closures	$K_0(z), D, K/K_0(m_d)$, capture, $P_{\text{esc}}(d_f/d_h)$	Solve flow and particle transport on CT-derived networks; export depth profiles and lookup tables.
Hydraulic experiments	global K scale, β_F , clean-basket R_s	Fit $\Delta p = c_1 Q + c_2 Q^2$ over several flows; low-flow slope constrains K , curvature constrains β_F .

These quantities should be fixed before fitting fines-release or detachment kinetics.

Dynamic experiments identify the remaining rates

Machine characterization

- free-flow and loaded pump curves;
- pump time constant τ_p ;
- flow cap $Q_{\max}$;
- PI gains and pressure target.

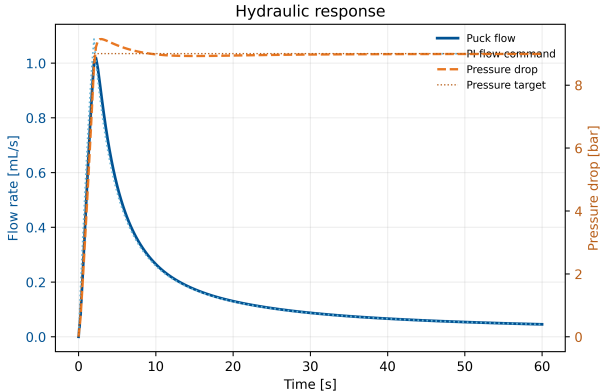
Fix these using known hydraulic resistances.

Fines dynamics

- early cup-fines curve $\rightarrow$ release;
- outlet PSD $\rightarrow$ size selectivity;
- interrupted CT/PNM $\rightarrow$ deposition location;
- pressure pulses $\rightarrow G_{\text{crit}}, k_{\text{det}}$;
- basket mass $\rightarrow \eta_s$.

Use multiple data types and validate on withheld pucks or grinding levels; do not fit every parameter to one pressure curve.

The illustrative case develops strong viscous blockage



At 60 s

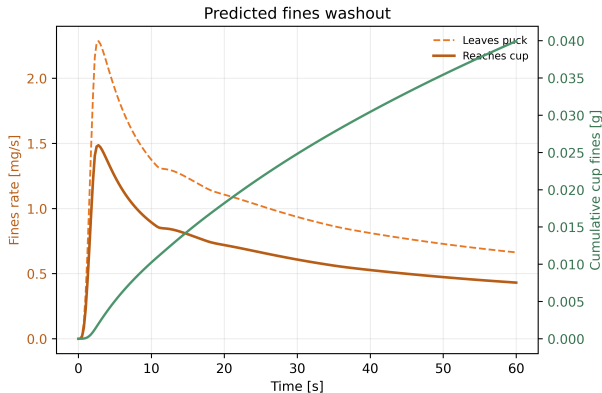
- total pressure: 8.99 bar;
- flow: 0.045 mL/s;
- minimum K/K_0 : 0.0126.

Interpretation

The PI controller approaches 9 bar while the increasing puck resistance strongly reduces flow.

Generated by the current Python solver; parameters are illustrative and not experimentally calibrated.

The model distinguishes puck washout from fines in the cup



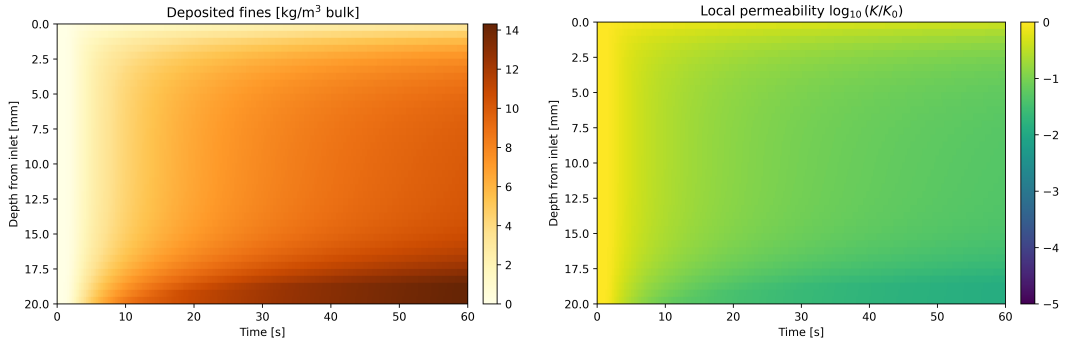
Cumulative masses

- puck outlet: 0.0614 g;
- basket retained: 0.0215 g;
- cup: 0.0399 g.

With $\eta_s = 0.35$, the basket retains 35% of the fines that leave the puck.

Generated by the current Python solver; level-1 basket retention is constant.

The axial fields locate the effective blocking region



These fields can be compared with interrupted CT reconstructions, but a 1D layer cannot represent lateral channel geometry.

Generated by the current Python solver; parameters are illustrative.

Verification and validation answer different questions

Verification: is the code correct?

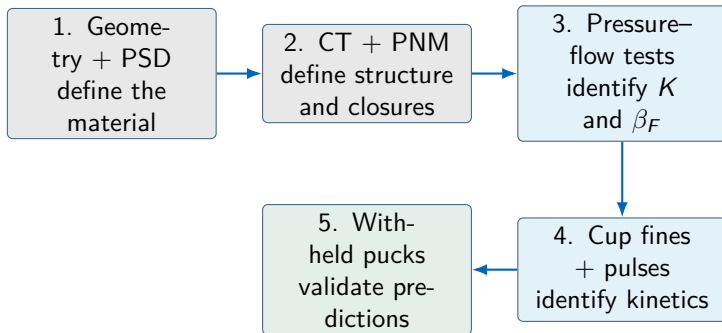
- exact Darcy and Darcy–Forchheimer solutions;
- exact pressure-to-flow inversion;
- conservation and positivity;
- non-oscillatory pure advection;
- grid and timestep refinement;
- PI and pump-cap limits.

Validation: is the model physically adequate?

- measured pressure and flow;
- CT-derived axial structural changes;
- PNM permeability and particle transport;
- cup-fines mass and outlet PSD;
- withheld grind settings or pucks.

Current status: 15/15 unit tests and 17/17 extended verification checks pass; physical validation remains the next stage.

The next experiments make the model predictive



A transparent 1D model becomes valuable when every closure is tied to an observable experiment or a pore-network calculation.

Immediate software extension: derive the fines fraction and representative diameters directly from the five PSD parameters.

References and model documentation

- M. Bora and H. Briesen, “Characterization of Bimodal Particle Size Distribution of Ground Coffee Powder,” *Journal of Food Process Engineering*, 49, e70645, 2026.
doi:10.1111/jfpe.70645
- S. Ergun, “Fluid Flow through Packed Columns,” *Chemical Engineering Progress*, 48, 89–94, 1952.
- FineFlow–Espresso technical report, Python solver, named-case configuration and automated verification suite accompanying this presentation.